

UNIT TITLE: SOEN 311: COMPUTER SECURITY

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REGISTRATION NUMBER: IN16/00032/24

ASSIGNMENT TITLE: ASSIGNMENT ONE

i. Highlight FOUR information or computing assets that require protection in the above case. (2 marks)

1. Student examination records
2. Examination marks and results
3. Course information
4. Examination Management Application and server infrastructure

ii. Identify TWO security threats and TWO vulnerabilities evident in the case. (4 marks)

Security Threats:

1. External attacker attempting unauthorized access by exploiting the known software vulnerability.
2. Compromised administrator credentials being used to access confidential records and alter examination marks.

Vulnerabilities:

1. Outdated software component with a known security vulnerability.
2. Examination administrators using weak passwords and possessing excessive access privileges.

iii. Discuss any THREE security controls introduced by the university to reduce the likelihood or impact of future security incidents. (6 marks)

1. **Updating vulnerable software:** Resolves the known security flaw in the system, directly reducing the likelihood of an external attacker exploiting the application.
2. **Introducing multi-factor authentication (MFA):** Enforces secondary verification during login, reducing the likelihood of unauthorized access from weak or stolen administrator passwords.
3. **Restricting administrator privileges:** Applies least privilege access, reducing the impact of compromised accounts by preventing unauthorized modifications to marks and examination results.

iv. Using the Examination Management Application case above, explain the following three fundamental security principles, giving an example from the case for each. (6 marks)

1. **Confidentiality:** Ensuring confidential student examination records and marks are protected from unauthorized access or exposure by external attackers or compromised administrator accounts.
2. **Integrity:** Ensuring examination marks, grades, and records cannot be altered, falsified, or tampered with without authorization.
3. **Availability:** Ensuring the examination management application remains operational, accessible, and running without disruption or downtime caused by software exploits.